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Connecting Enterprise Architecture and Portfolio Management

A systematic literature review

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Daniela Wolkerstorfer

Registration Number 12122373

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Advisor: Prof. Dr. Henderik A. Proper

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Daniela Wolkerstorfer

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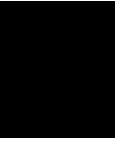
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Abstract

Enterprise Architecture focuses on a holistic overview and coordination within an enterprise, specifically addressing their respective technical infrastructure and the associated business processes. The increasing globalization has led to a gain in importance of this research area over the last two decades. Portfolio Management on the other hand has been an established management technique for an extended period of time. As the structure of enterprises become increasingly complex, the relevance of portfolio management has correspondingly intensified. The increasing significance of both management techniques, as well as their shared objective to provide a holistic and well-structured overview, substantiates the rationale for establishing a collaborative integration between them. In this thesis, the current state of cooperation should be examined, especially focusing on the potential benefits that a strong connection can yield across various fields. Furthermore, it aims to encourage the development of such connections if not present already, as well as to adapt strategies which have thus far been primarily theoretical in literature.

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Introduction

New technologies and business models, more complex corporate structures, globalization and the trend towards a service-oriented economy are aspects that many companies increasingly have to face [13]. As an approach to effectively cope with these arising challenges, the field of Enterprise Architecture (EA) has emerged [29]. However, EA is not to be seen as a completely new and stand alone field [29]. Instead, its aim is to provide concepts to expand traditional approaches like strategic or programmatic management [29].

Another field which has been emerging rapidly over the last decades is Portfolio Management (PM). Companies continuously become more projectified [17] describing “*an increasing reliance on temporary organisations, typically projects, in order to enhance action strategic effort*”[12]. This leads to the increased need to keep a good overview over all components and their according budget and strategic focus [17]. To cope with these challenges, Portfolio Management serves as concept to generate a holistic understanding to effectively manage all portfolio components [17].

Before being able relate these fields which might seem like individual and independent at first, a comprehensive understanding of these terms should be reached.

To attain such comprehension one can firstly disassemble each term into its individual components. Starting with EA one gets “Enterprise” and “Architecture”.

According to The Open Group Architecture Framework (TOGAF) [45] an Enterprise describes an organization on its highest level with all its missions and functions. Often one enterprise consists of more than one organization. This definition is relatively generic. Therefore, one can think of an enterprise if it comprises of a least one organization together with business and technological aspects together with human capital [29]. With these resources a company is goal oriented [29].

The term architecture originates from the art of building [2]. However, it has been increasingly used in the IT sector starting from the end of last century [2]. This usage does not only describe the structure of the computer but also its logical foundation as well as its implementation [2]. Subsequently, architecture was also used in the area of software development [40]. TOGAF defines the term architecture based on the ISO/IEC/IEEE 42010:2011 description [45]. The latter stating that architecture describes the essential ideas or characteristics of a system in its surroundings as represented by its components, connections and the guiding notions of its creation and development [45]. TOGAF builds on that and defines it as “*The structure of components, their inter-relationships and the principles and guidelines governing their design and evolution over time*” [45].

The necessity to coordinate enterprises not only in terms of their business activities, IT infrastructure and software, but also in terms of personnel, organization and information led to the emergence of the term “Enterprise Architecture” (EA) to characterize this holistic approach [29]. However, as the definition of the individual words already leave a lot of room for interpretation, the combination is even more prone to different understandings. Saint-Louis et. al. [38] found more than 150 different definitions. In some of these definitions EA is described as a process leading to a certain output, while other use the term to describe the output achieved. Another major difference in understanding is that some definitions see EA as a way to obtain a comprehensive view of an enterprise while others describe a reductionist vision. Additionally, while some describe EA as producing well-defined organizational components, others emphasize the enormous complexity of an organization, which runs counter to the notion that all components may be well-defined. To exemplify these differences, specific examples from three major organizations in the IT domain are presented. Firstly, IEEE [42] defines EA as a technique for managing rising complexity within enterprises by assuring the optimal use of technical resources. EA provides a holistic view of an organization’s fundamental structure, including its people, processes and applications, as well as their interdependence. Furthermore, EA should be seen as an option to guide evolution. The Gartner Group [14] defines EA as a field that leads enterprise responses to disruptive forces in a proactive and holistic manner by identifying and analyzing change execution toward targeted business results. Already analyzing a few definitions and taking them into account, Op’t Land et. al. [29] conclude that EA is a collection of explanations addressing an enterprise from the perspectives of governance, design and patterns. These perspectives offer controls and indicators facilitating an informed management of the growth and success of the enterprise. All in all, despite the various approaches and understandings of the term enterprise architecture, there are some aspects all of them have in common. There is consensus that various parts of the enterprise are needed to work together and their interdependence is of importance. Moreover, EA should at least contribute to, if not drive, change and growth.

Just like Enterprise Architecture, Portfolio Management is defined differently in the literature. To establish a shared understanding, it is important to first familiarize oneself with the concept of a portfolio. According to the Project Management Institute (PMI) [19] a portfolio is an array which might contain programs, projects, or operations

that are coordinated for the purpose of achieving strategic goals. The Infrastructure and Projects Authority of the UK [18] adds other portfolios, additional work and work packages to the possible components of portfolios. However, it is not necessary for every component of the portfolio to be interdependent or to have related objectives. With elements from the past, present and future, portfolios help organizations to achieve their goals and strategies. Unlike specific projects which typically have a set starting and ending point, portfolios entail a long-term perspective where projects might enter or leave the scope of the portfolio. Furthermore, an enterprise might have multiple portfolios. These portfolios should generally reflect the goals, trajectory and progress. Consequently, PM is the strategic control and management of one or more portfolios with the goal of aligning them with organizational strategy and objectives. As there are only finite resources available PM selects, prioritizes and distributes these in order to effectively fulfill organizational strategies that are in line with the enterprise's vision, mission and values. Given the dynamic nature of strategic goals, the portfolios will change accordingly. Therefore continuous monitoring of internal and external changes are necessary to ensure alignment with evolving objectives. A similar approach is shared in the definition provided by UK's Infrastructure and Projects Authority [18] which put huge emphasize on the fact that effective prioritized is essential for corporate planning and control. Moreover, this definition argues that a balance of change and established routine activities is sought through project management. In contrast to the presented international definitions outlined previously, the German approach focuses on the instrumental character of Portfolio Management [47]. This perspective is defined in the German DIN 69909 [7] standard as the comprehensive planning and control of project portfolios, including management duties, management organization, management methodologies and management tools [47]. In conclusion, every definition emphasizes the fact that PM involves but is not limited to managing at least one portfolio. Further management is required as other critical aspects of an enterprise such as alignment with the overall strategy and future plan of the enterprise have to be considered simultaneously. Lastly, maintaining a comprehensive understanding of the current state as well as the changes are part of PM which leads to the dynamic nature of portfolio management.

On first sight, Enterprise Architecture and Portfolio Management might seem to be two individual and independent divisions within an organization. However, having explored various definitions, some common ground can be determined. Both of them aim for a more prone relationship between different components of a organization, especially focusing on handling the steadily increasing complexity of enterprises. Furthermore, they both focus on the dynamic nature of each field striving for the most efficient contributions for change. Logically, one could conclude, that the best results can be achieved when Enterprise Architecture and Portfolio Management work together towards their shared goals. However, as seen in the definitions neither part discusses the relation to the other component. This might lead to the two divisions working towards different objectives despite their synergies leading to potential struggles to achieve each individual but especially the enterprise's overall strategy.

Therefore, as the main beneficiary of a well coordinated and thought through alignment of EA and PM, the enterprise as a whole can be seen. In fact, all stakeholders can be affected, however the severeness might differ. Firstly, every employee involved in strategic planning and decision-making will notice the effects of poor or missing coordination. They may have to change the planning over and over again if other, possibly conflicting, objectives change. Furthermore, the decision-makers are inherently affected by this challenge. Decision-makers must recognize the conflicting objectives and take proactive measures early enough so that there are no negative effects that may impact the entire organization. In the worst case, such effects could jeopardize the enterprise and even threaten its survival [20]. Further, even if only indirectly, all other stakeholders such as customers, suppliers and, potentially, shareholders can also be affected. These issues are mainly caused by missing collaboration of these fields. Also, there is a lack of understanding of the importance and impact of working together in this regard. Even if each component is integrated very well in an enterprise, the integration of EA and PM together is lacking mostly. Especially, the quickly evolving fields regarding technology are prone to this problem and even worse consequences can be expected due to the rapid changes.

This paper aims to provide an overview of already existing approaches of literature on how EA and PM can be linked effectively. This should be done by discussing and combining various approaches to align these divisions. Therefore, the implementation process of both EA and PM individually is of interest. As soon as it's foundation is laid, further determination on where and how the integration can be targeted most effectively, can be determined. Furthermore, it is of interest how all stakeholders can gather a shared understanding to work towards a unified goal. For this purpose models illustrating the connection between PM and EA should be employed. This information should be gathered by facilitating a comprehensive literature review to lay the basis for obtaining a comprehensive understanding of both divisions. Building upon this, various approaches as well as proposals of models fostering the integration should be discussed. The models given should place the greatest emphasis on clearly displaying the linking points as well as effectively demonstrating the benefits of integration in order to produce the best possible comprehension and holistic perspective of the existing situation. This thesis, however, does not claim to propose any new concepts on how or when to facilitate the integration. The scope does also not contain the proposal of new modeling techniques to foster possible connections.

For the search process Google Scholar was used as an initial source. To ensure the inclusion of the most recent research, the "cited by" feature by Google Scholar was employed. For further research, ACM Digital Library, IEEE Xplore and Springer Link were utilized. The following keywords were used during the search process: "enterprise architecture portfolio management", "enterprise architecture and portfolio management", "connecting enterprise architecture and portfolio management", "ea project portfolio management", "portfolio management through enterprise architecture", "enterprise architecture through portfolio management" and "business alignment through enterprise

architecture and portfolio management”. Furthermore, to also capture research published in German, “Unternehmensarchitektur und Portfoliomanagement” and “Unternehmensarchitekturmanagement Portfoliomanagement” were used as search terms. Following the search process, the potentially relevant papers have been evaluated regarding their actual relevance. The initial inclusion and exclusion process was conducted by analyzing their abstract. Papers which appeared to be relevant for the research objective were thereupon read completely to further evaluate their suitability. Finally, 11 papers were selected for the analysis in this thesis. The main reasons for excluding a high number of research papers were irrelevance and duplicates. Table 1.1 summarizes the selection process.

Source	Initially Evaluated	Selected
Google Scholar	24	10
ACM Digital Library	6	0
IEEE Xplore	8	3
Springer Link	7	3

Table 1.1: Summary of literature search and selection process

As a target audience of this paper, executives and decision-makers in enterprises, especially those responsible for enterprise architecture and project management, can be identified. Furthermore, students and researchers should be provided with a comprehensive overlook regarding the integration of Enterprise Architecture and Portfolio Management.

The remainder of the paper is structured as follows. Before being able to focus on the various different approaches describing where EA and portfolio management have some connecting points chapter 2 provides an more detailed look into both domains to foster the complete understanding of the following chapters. Subsequently chapter 3 provides the proposed relationships and connecting points described in literature. These will be viewed from different perspectives. Firstly, focusing on the view of EA on PM. Subsequently the other way around, describing when Portfolio Management aims to work together with parts of Enterprise Architecture. Then, the very limited work on a real interrelation will be provided. Consequently the findings will be discussed in chapter 4 Finally, a summary and recommendation for further research will be provided in chapter 5.

Background

2.1 Enterprise Architecture

2.1.1 Key concepts of Enterprise Architecture

In order to effectively implement EA four main components can be determined which will be presented in the following.

Principles in general are defined as “*comprehensive and fundamental laws, doctrines or assumptions*” [16]. Op ’t Land and Proper [28] recognize the significance of architecture principals by defining them as being crucial to guaranteeing enterprise architecture effectiveness. However there is no generally accepted definition of EA principles. Greefhorst and Proper [13] define EA principles as normative statements which consequently guide, but also restrict, organizational design choices. Haki and Legner [16] describe EA principles as a set of standards that distinguish between desirable and undesirable behaviors, hence restricting freedom of action. These restrictions are implemented in order to meet certain requirements. TOGAF [45] on the other hand describes two types of principles, Enterprise Principles and Architecture Principles. The former serve as the foundation for decision-making processes within an organization, guiding it toward achieving a harmonized process. The latter demonstrates a level of agreement throughout the organization and describe the requirements and methodologies of working through business goals and drivers.

Combining these different views one can conclude that EA principles guide organizations through changes with certain restrictions being made. Moreover, principles are there to assess the undertaken actions.

Models in general are viewed as abstractions of reality [29]. The main objective of modeling is to communicate facts in order to both intuitively answer questions and to enable prediction of reality [30]. Models are universally understood to be graphical

representations, however this is not necessarily the case [29]. When modeling enterprise architectures a variety of models is needed to represent all aspects which might be of interest to the different stakeholders involved [30]. This variety involves multiple dimensions, such as levels of realization, aspects of transformation, enterprise components and levels of aggregation, each serving a distinct purpose within the architectural process [29]. A modeling language specifically designed for the representation of different dimensions is ArchiMate [26]. ArchiMate defines three layers, the Business, Technology and Application layer [26]. These layers can be examined individually but they also allow for the observation of interdependencies which is not attainable in other common modeling languages like UML or BPMN [26].

Views serve as a representation of an enterprise from various perspectives [29]. Providing different views is important because different stakeholders need varying degrees of complexity and details [21]. Some might only need information regarding a subdomain of the enterprise while others might need to see the whole enterprise with a high level of abstraction [29]. Therefore, different views are necessary to cater for the varying needs stakeholders might have to get a combined understanding [29].

Frameworks enable the structuring of various perspectives while modeling EA artifacts [30]. There are a wide variety of Frameworks for EA available.

The initial one being published by Zachman [48]. In its most basic version it shows the design artifacts that connect the various roles in the design process. Depicted are these artefacts in a matrix-like structure consisting of six rows and six columns. The rows hold information about viewpoints of different stakeholders namely Planner, Owner, Designer, Builder, Subcontractor and Worker while the column represents the different perspectives being What, How, Where, Who, When and Why. All columns are said to be of equal importance. As the resulting cells are quite generic, the framework can be implemented in nearly any enterprise. All in all, the framework was the first one being presented of its kind and it is very easy to understand, However, there is no specification on how the different cells might interact or interrelate.

TOGAF [45] on the other hand describes different Frameworks namely the Architecture Capability Framework, the Architecture Content Framework and the Architecture Development Method. The first one offers a collection of references on how to set up the proper organizational structures, procedures, roles and duties to realize Architecture Capability. The second one is developed to allow for a stand-alone use of the framework not needing any other Frameworks like Zachmans. It divides work items into three primary categories. Firstly deliverables which typically represent work outputs. Secondly, certain elements of the architecture called artefacts. And thirdly, building blocks which are definable sections which might be reused in other parts of the framework. Finally the Architecture Development Method, which can be considered as the core of TOGAF, should guide architects through their work. It is based on an iterative cycle consisting of different development phases. Within these phases, goals are set and the results should be aligned with the expectations throughout. As the focus of the Architecture Development Method is on the process, it can be used together with other frameworks such as Zachman.

Despite having guidelines and a very detailed standard describing Enterprise Architecture provided by TOGAF it seems like the concept is difficult to implement as a lot of enterprises fail to successfully apply EA [29].

2.1.2 Benefits

However, if enterprises are able to successfully implement EA, they report a number of benefits they experience because of that. Most commonly these benefits include increased responsiveness to change, better decision-making, better communication and collaboration, cost savings and improved business-IT-alignment [44]. Moreover, EA also plays a major role in creating competitive advantage and increasing value [39]. These aspects could be interpreted similarly, however, the main difference of these umbrella terms are that the first one is focused on financial benefits while the second one is not [39]. However it is not clear whether these added values are a direct result of EA or they are indirect benefits [44].

In a study conducted by Saleem and Fakieh [39], a broad overview of the benefits EA was presented. Starting with business agility which includes, but is not limited to, assisting with the adoption of new technology, efficient process of expansion into new markets, increase of the operational capacity and more efficient responses. Furthermore, the creation of a competitive advantage can be observed. However, the observed advantages might be directly or indirectly related to EA. Especially, when an enterprise first starts implementing EA there are a number of benefits which can be seen as a direct result of the implementation. First of all, there will be a lot more organized and readily available data about the enterprise's processes as well as those of its suppliers and customers [44]. This contributes to the growing competitive advantage by making it easier to monitor competitors and the current market situation [39]. Gathering all this data is a task which involves numerous different parts of the enterprise. [44]. While gathering the information increased understanding of different but interrelated parts of the enterprise will be built [44]. Moreover, one might find interrelations which nobody has thought about before [44]. This in turn can make the process of decision-making easier as potential conflicts can be targeted in an earlier stage of the process [44] and it also makes the process more efficient, reducing costs [39].

Furthermore, appropriate resource allocation and planning is also a factor to reduce costs [39]. EA contributed by revealing the resource portfolio meaning it shows how much resources are invested in each part [44]. This in turn, enables the identification of both underinvestment and overinvestment [44]. While the former can be identified more easily due to its impact on business performance, the facilitation of the latter is mainly enabled through EA [44]. For example, EA can reveal situations when similar but different systems are used across various departments, which could eventually be combined into a single system [44]. As also the current IT situation is evaluated in this step, an easier IT platform with reduced complexity as well as reduced maintenance and support costs might be suggested [44]. Finally focusing on increasing value. One part of that is related to gathering information about customers as deeper knowledge of the

customers allows for better customer service also increasing non-financial values [39]. Another benefit worth mentioning is, that a clear EA goes hand in hand with a specific identification of the future vision of the enterprise [44]. This allows for every employee to understand the overall goals and targets [39]. This in turn ensures that everyone has the same vision and knows which target they are working towards leading to a feeling of safety and satisfaction in working for a shared purpose [39].

2.2 Portfolio Management

2.2.1 Key Concepts and Goals

In order to effectively facilitate portfolio management there are some key concepts which have to be understood.

First of all, one should shed light on necessary **requirements**. There are three essential ones as Ershadi et. al. [8] argue. The first of them being structure which should provide a solid background for the government of arising portfolio management tasks. It is necessary to precisely define the responsibilities within a portfolio. In fact, it has to be clear how decisions are taken, which criteria influences the decision making process as well as which stakeholders play a role in this process and who takes and is responsible for the resulting decision in the end. Therefore, some parts of the portfolio will have a higher level of control than others. This goes hand in hand with the identification of which decisions are taken by which part of the portfolio. Some smaller ones might be taking place within a project or program without the need for approval of other parts of the portfolio, while bigger ones have to be supported and discussed throughout the whole portfolio. The second key resource is infrastructure which has to be available to successfully run portfolio management within an organization. This infrastructure mainly focuses on IT infrastructure to enable effective coordination, control, evaluation and visualization. The IT infrastructure must provide sufficient data necessitating the utilization of appropriate data collection tools, in well structured data bases along with suitable software applications. Thirdly, processes are identified as an essential resource for good portfolio management. In this context, processes are described as the needed actions to transition from the current state to an improved state. As previously discussed, portfolios are not static entities but undergo dynamic changes over time. Therefore, there has to be a structured process to firstly identify possible new portfolio components. Following, there has to be an evaluation and potentially an approval to add the new component. Also, the removal of portfolio components have to follow a structured process. These defined paths are necessary to make sure that only programs, projects or operations which align with the enterprises overall longterm strategy are added or kept in the portfolio.

To account for those changes the whole **life cycle** of a portfolio has to be considered. In the Standard for Portfolio Management [19] four stages have been identified. The first one being Initiation entailing the previously described key requirements. As such, the

management strategy, the components and long term goals and roles are defined. Afterwards the planning stage is reached. In this stage, awareness has to be formed overseeing the various parts of a portfolio, which includes budgeting, risk management, identifying connections between components, determining the required resources, prioritizing within the portfolio, designating a key person for oversight, monitoring progress, defining what is included in the portfolio and identifying which goods or services are required. Having defined all of those aspects the execution phase will follow. The goals are to check the delivery of each portfolio's components, manage occurring risks and issues within a portfolio as well as interconnected parts, facilitate effective communication, if needed reprioritize certain components, track benefit realization and manage the resources and assets specifically assigned to the portfolio. Lastly, optimization will take place with the primary goal to ensure that the resource allocation is done effectively without over or under allocating. Although this process seems like it always has to follow this specific structure, in some cases this life cycle might be messed up because several planning stages might be needed. Moreover changes might need to be made if the overall strategy of an enterprise changes, leading to a mix up in the life cycle.

A **framework** to assess portfolios must at least consist of the information needed for the life cycle. This means information on how to select the appropriate portfolios, how the allocation of resources should be targeted, how to align the portfolio with the overall strategy and how success can be measured [32]. The framework has the goal to allow for rational and transparent decision making [32]. Common methodologies are provided by the UK Office of Government Commerce or the Project Management Institute (PMI) [45]. The first one presented "Projects In Controlled Environments" (Prince2) which is built upon experience as well as expert knowledge and is primarily used in the UK [33]. It focuses on the business strategy and aims to provide an organizational structure for the project management team prioritizing the desired outcome and breaking the project down into phases [33]. PMI on the other hand presented the Project Management Body of Knowledge (PMBOK) which main objective is to guide project managers successfully fulfilling the goal of a project taking the whole project life cycle into account [33].

Altogether four main **goals** of portfolio management can be identified First of all, maximizing the value of the portfolio, accounting for the given constraints like financial factors, risk awareness and limitation of resources [4]. Secondly, finding a proper balance of projects within the portfolio which might look like a mixture of long term or short term components and elements with high versus low risk [4]. Moreover, as already discussed, the strategic alignment of the portfolio not only has to be in line with the overall strategy but should completely reflect it in each aspect [4]. This has to be the case at all times [31]. Finally, an appropriate number of components has to be selected as there is only limited resources [4]. A new project can only be incorporated if sufficient resources are available to sustain progress within the already existing components and to achieve the set goals of the potential new project [4].

2.2.2 Benefits

With these goals in mind, one can determine the main benefits an enterprise can get from a well set up and thought through portfolio management. As described by Killen [24] portfolio management is a key aspect to achieve organizational agility which helps to enable competitive advantages. Competitive advantage refers to an organization's capability to generate more value than its competitors, resulting in a greater return on investment. Agility is especially important if dynamic responses to environmental changes are necessary. Given the rapid pace of change in dynamic environments, continuous learning is necessary to ensure that the whole project management process remains up to date. In fact, portfolio management enables learning on two levels. Firstly, tactical learning which is acquired through experience or observation during portfolio managers' meetings. Secondly, explicit learning which represents formalized knowledge clearly documented in items like templates or databases. This continuous learning aspect provides competitive advantage over other enterprises. Another major advantage of portfolio management which also becomes especially present in dynamic environments is appropriate resource allocation. In quickly evolving contexts, certain programs, projects or operations may appear highly promising at first. However, this may change due to environmental changes such as the advent of a new product or technology, resulting in varying success proposals for a component. Without functioning portfolio management stakeholders tend to be resistant when it comes to the thought of cancellations or major alterations within the portfolio. However, through its assistance, allowing for transparent criteria and data, a better understanding and support can be reached. Furthermore, the ability of an organization to effectively exploit and explore simultaneously is crucial in many sectors. Exploitation is typically short-term, low-risk and incremental which is necessary to improve current services or capabilities. Exploration on the other hand, is characterized by long-term, high-risk and radical projects aimed to generate innovative breakthroughs. Portfolio management can be used by an enterprise to ascertain the balance of these project types to compile data across the portfolio.

Another aspect where portfolio management may positively support an enterprise is when targeting innovation [23]. Because of shorter product lifecycles and more dynamic markets, the development of new products is becoming more and more important in today's dynamic corporate climate [23]. Portfolio management therefore strides to increase success rates when developing product innovation by providing an all-encompassing environment for decision-making [23]. In fact, correlations between adequate portfolio management and improved results in new product development can be observed which in turn again contributes to a competitive advantage [23].

Accounting for the wide variety of areas of application provided by PM, several distinct subdomains can be identified [22]. Relevant for these thesis are Application Portfolio Management, Technology Portfolio Management, Transformation Portfolio Management, Project Portfolio Management as well as IT Portfolio Management. The connection each of the listed subdomains to EA will be discussed in the following chapters.

2.3 Overview

Altogether, despite their distinct approaches and focal points, Enterprise Architecture and Portfolio Management seem to share some commonalities. Some of their goals overlap, however their pathways might differ. Following the explanation of the concepts, it appears that collaboration between these two fields could benefit organizations. To create a comprehensive understanding of the different approaches taken by Enterprise Architecture and Portfolio management table 2.1 should present an overview and juxtapose the key points.

	Enterprise Architecture	Portfolio Management
Basic idea	Interdependence of various parts, driving change and growth	Managing at least one portfolio, dynamic nature
Key concepts	Principles, Models, Views, Frameworks	Requirements, Life cycle, Frameworks
Key Frameworks	Zachman, TOGAF	PRINCE2, PMBOK
Applications	Decision-making, Business-IT-alignment, Resource management	Dynamic environments, Resource allocation, Innovation
Benefits	Responsiveness, Communication, Cost saving, Competitive advantage, Increasing value	Learning, Organizational agility, Competitive advantage

Table 2.1: Summary of Enterprise Architecture and Portfolio Management

Intersection

Enterprise Architecture and Project management both aim to benefit the business processes by providing structured approaches on how to achieve the best possible outcome. This chapter should therefore provide a comprehensive look into the positive results which can be archived by integrating both management methodologies within one organization.

3.1 View of Enterprise Architecture on Portfolio Management

TOGAF [45] provides a Architecture Development Method (ADM) which allows relating EA with other frameworks, project and portfolio management being one of them. To get a first overview one can see the proposed relationship by TOGAF [45] depicted in fig. 3.1.

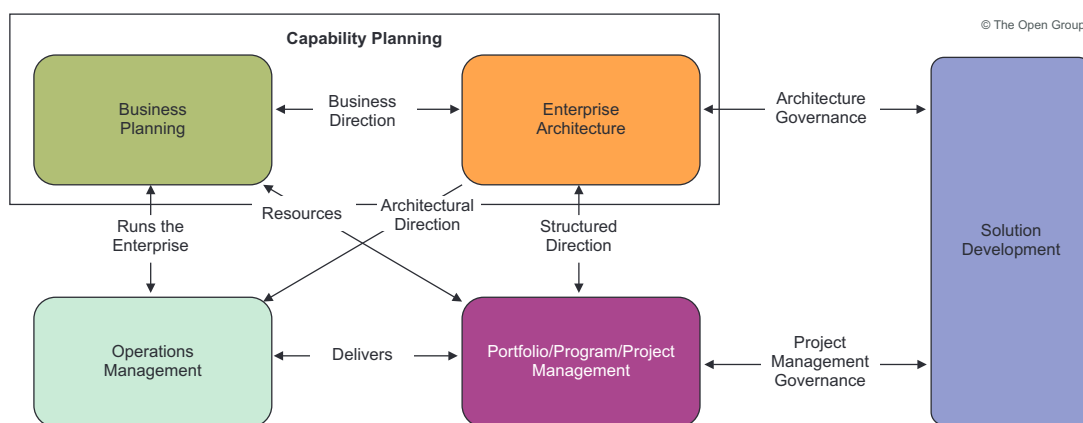


Figure 3.1: Relationship between Frameworks [45]

TOGAF argues for a structural direction between EA frameworks and PM frameworks. This should provide a well structured guidance allowing for ensuring that each portfolio is aligned with the overall enterprises' goals.

However, the reader should be advised that this reflects the viewpoint of an EA standard and does therefore not necessarily constitute a comprehensive review.

3.1.1 Architecture Development Method

The ADM can be divided into eight phases with the goal to “*develop and use an Enterprise Architecture to shape and govern business transformation and implementation projects*” [45]. Connections to portfolio management can mainly be found in the later phases E to H. Figure 3.2 depicts the cycle which will be explained in the following, mainly focusing on the later phases.

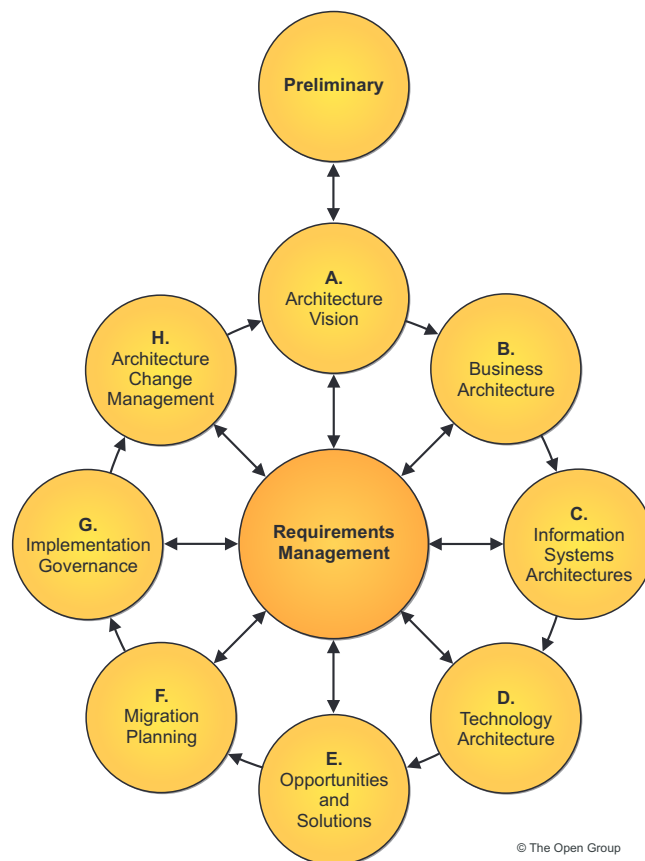


Figure 3.2: Architecture Development Cycle [45]

Phase A, also referred to as the Architecture Vision phase, should define an objective the architecture should achieve. For that, a clear definition of aspects which are within

the scope but also specifically which ones are out of scope are specified. Moreover, stakeholders and their interests have to be discussed and approvals need to be obtained.

The second phase, Business Architecture, should provide a to-be Architecture and also identifies the steps required to reach this target.

Thirdly, Information Systems Architectures, phase C, adds Information Systems to achieve the determined to-be architecture from phase B.

Subsequently, phase D, called Technology Architecture, determines the needed technology to implement the enterprise architecture and again presents the actions needed to be taken to reach the desired technology state.

Phase E, Opportunities and Solutions, uses all the information gathered in the previous steps and aims to determine the steps which have to be taken in all aspects to reach the vision architecture. This should result in providing and aligning “*projects, programs, or portfolio*” [45] as clearly stated by TOGAF.

The proposed “*projects, programs, or portfolios*” [45] are subsequently migrated into the current situation in Migration Planning.

Phase G, called Implementation Governance, aims to give an overview regarding the work performed. Here it is important to again verify the conformance with the previously defined to-be architecture.

Finally, in phase H, Architecture Change Management to create protocols to oversee modifications to the new architecture and manage changes throughout the whole life cycle.

3.1.2 Resource Portfolio Optimization Through EA

Investing in a well-defined EA goal and implementing the thought of architecture comes along with various improvements across different areas of the enterprise as well. One of them is resource portfolio optimization as discussed by Tamm et. al. [44]. The resource portfolio of enterprises can be described as the collection of “*all assets, capabilities, organizational processes, firm attributes, information, knowledge, etc. controlled by the firm*” [3]. In the context of enterprises, resources are mostly IT, human resource and business processes. All components of this portfolio have to be distributed to provide every division with the needed resource without overinvestment or underinvestment. While planning an EA, which is covered by phases B to E in the architecture development cycle, the areas of overinvestment or underinvestment will become obvious. Taking a closer look into the used IT resources yields the opportunity to build a standardized IT platform where every division of the enterprise works with the same technology. Furthermore, the process allows for the specification of the used components, which provides the opportunity to change components of the resource portfolio more easily as needed. This can help to simplify processes resulting in higher reliability, lower complexity and reduced maintenance and support needs. Moreover, the documentation which comes

along with this process reduces the future risk of replacing components and provides awareness about each component and their interdependencies to other parts. All in all, EA mainly helps to reduce redundancy and consequently costs. This benefits are most pronounced when a big enterprise with a lot of dependencies and high complexity first start to optimize their resource portfolio. In fact, the worse the initial overinvestment and underinvestment the more value an EA will add to the enterprise.

3.1.3 EA enabling Application Portfolio Management

Also related to generating the optimal resource portfolio is the selection of the needed application portfolio which also serves to avoid and uncover redundancies [25] [34]. An application component is defined as follows by ArchiMate: “*An application component represents an encapsulation of application functionality aligned to implementation structure, which is modular and replaceable*” [46]. Furthermore, TOGAF states that it is preferable to design applications which are used throughout the entire organization as opposed to duplicate or comparable applications being exclusively available to certain divisions [45]. Therefore the selection of the appropriate application components plays a major role in EA and the definition of an application portfolio is often a starting point of EA initiatives [25]. These initiatives will usually lead to changes in the application portfolio which means that new innovative applications might be implemented or the current applications might have to be changed [25]. Consequently, documentation is a crucial part of application portfolio management (APM) since it allows you to keep track of every component and determine which ones are inventive, long-lasting, or, in some cases, short-lived [25]. The evaluation of the investment strategy and life cycle, the analysis of information flows and interfaces, the adaptability to user needs, the application of architectural principles, and the assessment of applications in terms of costs, stability, and technological fitness with corresponding recommendations for measures are additional aspects that should be covered by APM [25]. Relating these objectives to the ADC leads to the conclusion that APM will mostly take place in phase C, Information System Architectures, where the alignment of business functions and applications is determined [34]. Later in the process of change, also phase E regarding solutions and phase H, Architecture Change Management, are associated with APM [25].

Inspired by the ADM, application portfolio management can be seen as a cyclical process model being a continuous process along with the overall enterprise architecture management [25]. A model of the process is depicted in 3.3.

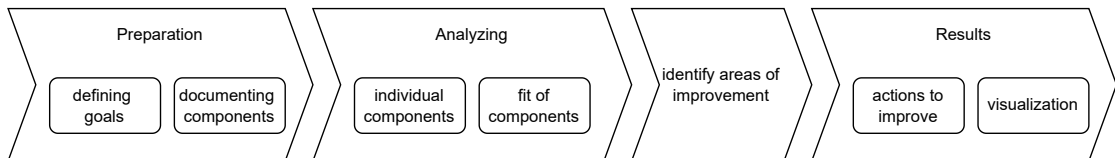


Figure 3.3: APM process (inspired by [25])

The first step can be seen as the preparation phase which consists of defining goals [15] as well as documenting the currently used components specifying their use and state of life cycle [25]. Moreover, also new recently introduced applications should be listed [34]. In this step, only the most important general information and information needed to identify potential for improvement should be gathered [25]. The information gathering process can be done by either automatically reading source code or databases, manually by assessing surveys or questionnaires or semi-automatically [34]. These activities could also be divided into two separate steps [15], however, as both of them help to provide a deeper understanding of the current landscape and future vision combining them into one preparation phase can be justified. The next step consists on analyzing the determined components [15, 25]. The valuation of each component will be based on several criteria which is individual for each enterprise [25]. However, some general factors to be determined are functionality, costs, safety and stability [25].

An example evaluation is provided in table 3.1. The first four applications are absolutely necessary for the enterprise and their cannot be questioned. The next two are newly emerging applications which might offer huge benefits for the enterprise which is why a closer look into these technologies is advisable. Finally, the last listed example should be carefully evaluated to determine why this application is still in the portfolio, since this system causes high costs but has no other added value according to these assessment criteria.

Application	Use	Functionality	Cost	Security	Stability	Life Cycle
Project Management Tool	Daily Operations	High	High	Medium	High	Mature
Accounting Software	Daily Operations	High	High	High	Medium	Mature
Database Server	Critical Infrastructure	High	High	High	High	Mature
Email Client	Daily Communications	High	Low	High	High	Mature
AI-Powered Analytics	Strategic Decision Making	High	High	High	High	Emerging
Cloud-Based Collaboration	Collaboration	High	Medium	High	High	Emerging
Legacy ERP System	Historical Data Management	Low	High	Low	Low	Declining

Table 3.1: Analysis of Applications

When assessing the current and future value contribution of an application component to the enterprise's performance, either the enterprise's goals or its strategic competencies must be considered [25]. Additionally, not only the valuation of each individual component is of interest, also the fit of the components comprising a portfolio should be evaluated [34]. Coming back to the example provided in table 3.1, legacy ERP system might be needed

in the big picture which cannot be determined by valuing the application on its own. In fact, only when considering all components, constraints concerning resource and costs can be holistically respected [34]. Depending on the quality and quantity of available data, multiple well-fitting portfolios can be calculated, leaving decision-makers with various options to weigh in the overall context [34]. Following the valuation, areas of improvement can be identified by comparing the results [25]. As a forth and final step, actions aiming for improvement can be set [25]. These could include the migration of an application component [25] which leads to phase F of the Architecture Development Cycle. If components do not fit well, they might be eliminated or tolerated depending on their corresponding costs and costs to exchange them [25]. In this phase, also a visualization of the results and possibly the actions needed to be taken to get to the proposed portfolio is indicated [15]. The mostly used and highest rated tool by the Gartner Group [10] for this is Orbus which also supports ArchiMate as a modeling language.[43]

3.1.4 EA enabling Technology Portfolio Management

Having defined the appropriate applications for the enterprises, the following step is to determine the best suited technologies to use within the components [9]. Technologies are in the narrow sense “*the use of scientific knowledge or processes in business, industry, manufacturing*” [6]. In a broader sense, the term can also include machinery and equipment [9]. This activity corresponds to phase D in TOGAF’s Architecture Development Cycle. Similarly to the previously described process of selecting applications, the final decision has to be completed no later than in Phase E [9].

Frenzel and Höllwieser [9] describe how technology portfolio management plays an important role in the creation or further development of an enterprise’s enterprise architecture. The main objective should be to only use a minimum of different technologies aiming to remove redundancies and consequently improving maintainability as well as reducing costs. Generally, the whole process to select a suitable technology portfolio is comparable to the process of choosing the applications and is schematically depicted in fig. 3.4.

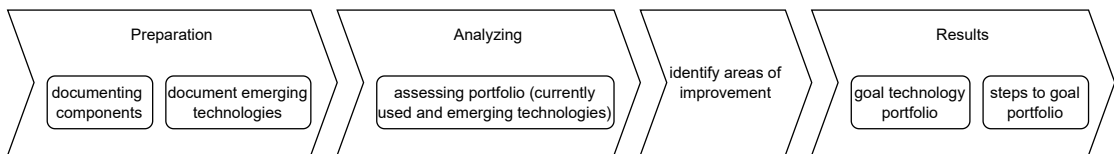


Figure 3.4: Technology portfolio process

Firstly, the current situation must be recorded. To do so, all technologies can be categorized into their respective application area. Within these areas, the level of detail can be varied accordingly. For example, software development can be divided into frontend and backend development. Consequently, frontend development will use specific technologies like HTML and CSS, JavaScript, Vue and React. The technologies used must

be regularly assessed as part of technology portfolio management. Alterations might occur due to adaptations within the organization or if there are modifications from a technology provider, for example in terms of support policies. Therefore, a detailed portfolio, with further information such as the status in the life cycle, version and manufacturer, is essential.

Furthermore, emerging technology must be observed and where applicable, studied in detail. Similarly to application portfolio management, the new technology must be evaluated regarding their overall improvement possibilities. In context of technology management, the knowledge the person responsible for the observation of the market must have greater in depth knowledge about the subject matter.

Having observed the currently used and newly emerging technologies, the assessment of both categories should follow. Concerning the former, new releases have to be monitored. Using outdated versions or ignoring patches can be a security issue. However, applying newly released versions immediately might also not be recommended as there might be unexpected bugs or compatibility issues. Evaluating the latter mainly focuses on their possible fit within the organization's portfolio. This also includes the consideration whether human resources within the enterprise exists to work with the newly emerging products. In fact, the rapid emergence of a large number of technologies leads to challenges maintaining an overview without a structured approach [35].

A possible assessment in line with the previous example concerning frontend technologies is represented in table 3.2. The first four are technologies which are safe to use and their use can be recommended. Newer versions should be evaluated. Underneath, a technology which should be evaluated for potential future use is registered. Lastly, AngularJS should not be in the portfolio anymore due to its end of life valuation. Usages of it have to be reconsidered.

Technology	Use	Functionality	Cost	Maintainability	Life Cycle	Version
HTML/CSS	Frontend Development	Styling and layout of web pages	Low	High	Mature	CSS3
JavaScript	Frontend Development	Interactive elements in web applications	Low	High	Mature	ES6
React	Frontend Development	Building user interfaces and single-page applications	Medium	High	Growth	16.x, 17.x
Angular	Frontend Development	Robust framework for building scalable web applications	Medium	High	Growth	10.x, 11.x
Svelte	Frontend Development	Innovative approach to build reactive web apps with less boilerplate code	Low	Medium	Emerging	3.0
AngularJS	Frontend Development	Framework for building scalable web applications	Low	Low	End of Life	1.x

Table 3.2: Analysis technology portfolio

As a result, a goal technology portfolio is determined which should be implemented in the following steps of the Architecture Development Cycle. The main implementation will take place in phase E as the application portfolio management, which partially relies on these technologies, needs to be finalized in this phase as well like previously outlined.

3.1.5 EA enabling Transformation Portfolio Management

Closely linked to the previously discussed portfolio management disciplines, transformation portfolio management serves to initiate the implementation of new transformation projects [36]. The capacity to quickly and efficiently restructure the enterprise to align with the evolving value offering is crucial in today's rapidly evolving economy [27]. It is also closely related to resource portfolio management as new proposition tend to be more risky than well-established ones [36]. As these activities mainly lay in the planning and initiation within the Architecture Development Cycle, preliminary and Architecture vision are the aspects affected [36]. However, further concerned are phases B C and D as there might be new ideas and proposals of new transformation projects after a detailed look into the different architectures [36]. As already discussed in the previous management disciplines, the adaptations are consequently made in phase E [36]. Moser and Brandmayr [36] argue, that one transformation initiative always involves a cycle through an ADM process. Along this process, at least but not limited to one project should be realised. fig. 3.5 represents the main steps within the transformation portfolio management.

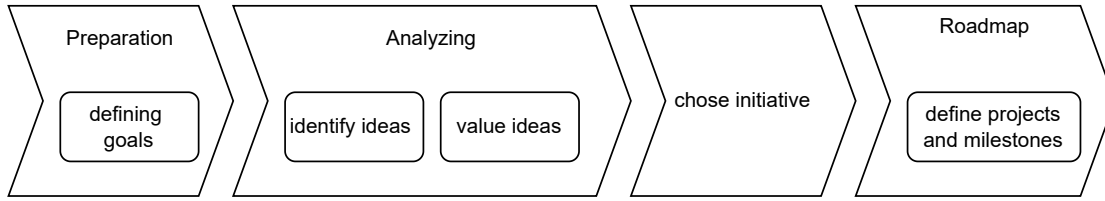


Figure 3.5: Steps for transformation portfolio management (inspired by [36])

The first step has to be to precisely describe the goals an enterprise is working towards. These not only need to be in line with the organizations overall strategy and goals but also have to take all regulations, political developments and future trends into account. In order to be able to agilely adapt to the fast changing environments, the process which will be described in the following should be carried out at least once every quarter. To achieve this effectively, the entire organization must be receptive to adaptations, and budget planning should remain flexible rather than being rigidly fixed for extended periods. Subsequently, all proposed initiatives will be collected and compared estimating their possible contribution to future corporate success. To gather as many ideas as possible, collecting ideas from the perspective of different teams is beneficiary. On the other hand, outsourcing idea generation to a specialized department is not recommended [27]. Subsequently the ideas have to be valued. For this valuation different evaluation criteria like price, resources and risk can be taken into consideration. The most promising ideas will be taken into the next step to further assess their fit and value contribution for the users or the enterprise. The assessment of the value contribution should be based on previously defined Key Performance Indicators which are necessary to select the initiatives but also later to have a clear plan how to measure the value contribution. Finally, a comprehensive plan specifically stating each project and milestones of the transformation initiation each specifying their respective value contribution must be defined. These are also necessary for resource portfolio management to allocate human and financial resources effectively. For a structured oversight of this whole process ArchiMate can serve as a modeling language.

3.2 View of Portfolio Management on Enterprise Architecture

3.2.1 Choosing appropriate components

One major part of portfolio management is making sure to only consider appropriate projects to enter the portfolio. Aldea et. al. [1] presented a methodology representing how EA can complement project portfolio management in providing a detailed overview about interdependencies within a portfolio. Moreover, it should facilitate the analysis of the resulting effects, costs, risks and benefits arising from the execution of a project. ArchiMate, as the main modeling language in Enterprise Architecture context, already provides a strong foundation in recognizing the impact a change in a portfolio might have, as architectural elements can be added or removed easily.

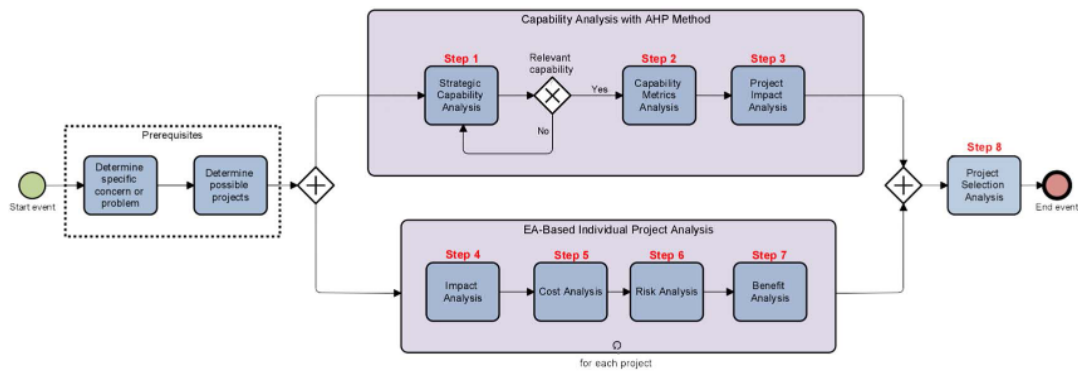


Figure 3.6: Project Selection Approach [1]

Before EA aspects come into play when choosing appropriate portfolio components some foundation has to be laid by defining capabilities. The whole process is depicted in fig. 3.6. The thought of fundamentals are represented by the initial three steps, which involve identifying the most influential capabilities as well as the metrics used for their definition and evaluation. These steps serve as the basis for selecting the appropriate components. Having defined these aspects, EA comes into play. Displayed in step 4, the interdependencies as well as the impact of a new project's implementation should be comprehended. This involves identifying changes across various architectural components to gain a comprehensive understanding. In this step, the future architecture is defined and all impacts these changes come with are evaluated. Subsequently, caused by changes in the Enterprise Architecture, also the cost structure has to be adapted which is depicted in step 5. Therefore, an analysis is performed in every part where the determined future architecture differs from the currently existing one. Following these adaptations also the resulting risk of EA changes have to be determined as risks within a portfolio are dependent on each component. As a final step in the scope regarding EA, Step 7 analyzes the benefits brought about by the addition of the project. To quantify the total benefit, each advantage is assigned an importance level. The determination of the

advantages provided by the addition is based on beneficial architectural changes. Steps 4-7 are iteratively gone through for all possible components which might be added to the portfolio. Finally, in step 8 the whole analysis is used to choose the best fitting component for a specific portfolio taking all previously collected data into account.

Evaluating IT portfolios using EA

For many enterprises, of special interest is the valuation of their IT artefacts [37]. The reason behind being the quickly evolving nature of the field [37]. However, enterprises only invest 10 percent of the budget designated to IT in innovative artefacts while the vast majority is used to maintain the already existing and unchanged IT landscape [37]. Striking the goal of a well-balanced portfolio Quartel et. al. [37] propose an approach specifically targeting the valuation of IT components also taking business processes into account. To establish this connection, AchiMate is used for modeling purposes.

According to Quartel et. al., [37] in IT portfolio management, the long-term existing landscape is typically referred to as application portfolio while the innovative components are grouped together under the term project portfolio. For a project to be added to the project portfolio, they have to be evaluated independently of other components first. However, the composition of those portfolios is not static. The process of the evaluation is represented in fig. 3.7.

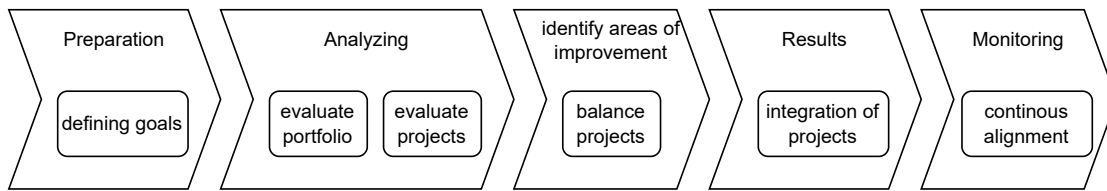


Figure 3.7: Steps for IT Portfolio Management and EA Alignment

Firstly, the artefacts of the application portfolio is supposed to be evaluated. To do that the overall strategy of the enterprise must be clearly defined. This evaluation is supposed to be the basis of further decisions whether a component of the project portfolio might be a good suit and should therefore be removed from the project portfolio but added to the application portfolio. Projects are therefore evaluated whether improvements in certain areas, like reduced risk or increased effectiveness are possible. The components are supposed to be balanced in terms of cost benefit and risks. This process results in a new, better aligned IT landscape and has to be monitored and eventually adjusted. Especially the alignment part is of interest when discussing the role Enterprise Architecture plays in IT portfolio management. In fact, EA serves as a tool to comprehend the relationships inside a portfolio providing the possibility to describe all three, the relationship between IT components or business artefacts as well as between the two. It should be possible to determine the appropriate level of IT support for an enterprise as a whole but also a more detailed look into the IT needs of business processes should be possible. To achieve that the effectiveness and the importance of IT are calculated against each other

aiming for balance. If for example, the importance is very high, also the effectiveness of IT is supposed to be very high. A change in the components utilized should be taken into consideration if there are significant deviations. Here, the project portfolio comes into play again as these pre-selected projects can all be valued and the best fit can be determined. This leads to eventual changes of the application portfolio and the adding of components of the project portfolio as described. Gathering all this data and valuating the possible components serves as an way to make the decisions taken transparent and understandable. Moreover, it provides a precise and systematic approach to choose appropriate components in a quickly involving field when also taking into account dependencies to other parts of an enterprise.

In order to model this alignment an ArchiMate extensions is presented by Quartel et. al., [37] which focuses in the implementation and migration for TOGAF'S architecture development method phases E to H. To the core of AchiMate concepts for modeling implementation projects and programs are included in this extension to facilitate portfolio, project, and program management. The connection between the core elements and the proposed extension is depicted in fig. 3.8.

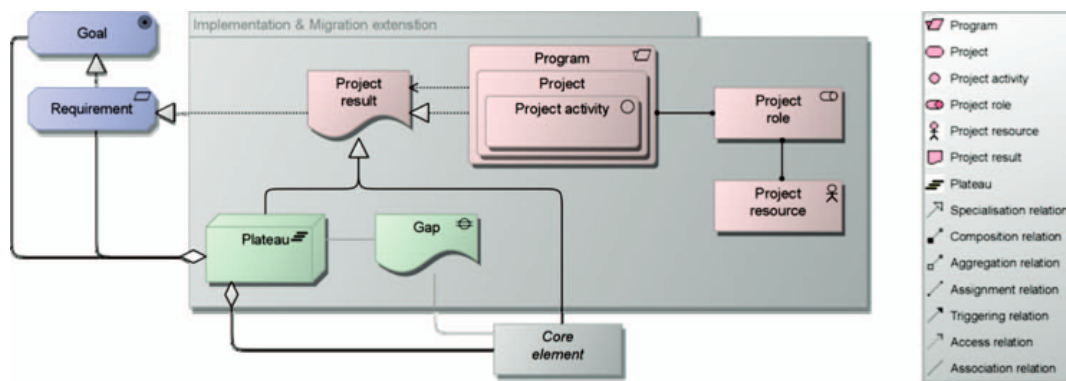


Figure 3.8: ArchiMate Extension [37]

As a base the requirements, goals roles and resource have to be defined. These are essential to compute the appropriate components for the IT landscape. The plateau is needed for the migration of the core and the extension. Additionally, the modeling of programs, projects and project activities is being made possible through this extension. Combining all of these aspects, a good result benefitting the enterprise and accounting for various needs different parts of an enterprise might have when it comes to IT and its interrelation with the business processes can be modeled.

Having these models and using the described techniques the portfolio with the best value and effectiveness can be determined. However, one has to keep in mind that some aspects like social influences and politics are not represented in these models. Therefore, the management team might take different actions than the approach suggests.

The described approach has significant similarities to the process discussed in section 3.1.2.

However, the main difference is the starting point. In section 3.1.2 the reason starting the whole evaluation is the initiation of an EA initiative leading to the necessity of an in depth look of the application portfolio of the enterprise corresponding to step C in the ADC. The approach described here is based on the need to be able to not only cope with the quickly evolving environment but also use it to the enterprises advantage. Therefore, the main focus is not only a thorough evaluation of the components but also, and even more pronounced than in the previous approach, to investigate the connections to other parts. Furthermore, the implementation and migration process is discussed in detail from the view of portfolio management which is has not been done by the research conducted from the EA view.

3.2.2 Governance

Within projects, programs or portfolios (PPP) appropriate resource allocation as well as coordination and control of the undertaken activities is directly related with a positive outcome of those [5]. A model of these actions in combination with the decision making process to maintain organizational values when changes are made can be combined in the term governance [5]. As this needs exactly defined roles, responsibilities as well as performance criteria, Cordeiro et. al. [5] use ArchiMate as their modeling approach for architecture models bringing Enterprise architecture into play. Roles in portfolio management should refer to Portfolio Manager or Portfolio Management Team as defined in ISO 21504. Cordeiro et. al. [5] present an approach which should act as a template to compare traditional project management and PPP governance models to allow for selecting the best suiting one for a specific enterprise. This template is a reference architecture divided into seven layers. The layers are Competences, Roles, Business Governing Layer, Steering Layer, Directing Layer, Managing Layer and Performing Layer - see fig. 3.9.

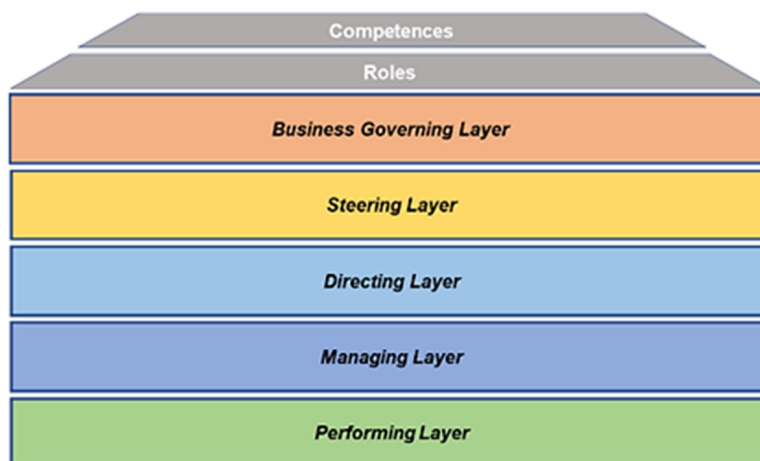


Figure 3.9: Layers [5]

This layered structure allows the definition of clear relationships between the function

and their roles and enable a clear link to their corresponding position in the processes and structures of the organization. Subsequently the scope of each role is analyzed with the viewpoint of the ArchiMate Business Layer.

Having specified the roles, each role will be associated with the previously described layers. This mapping can be modeled by ArchiMate and becomes possible because Enterprise Architecture allows for relating the identified role to their corresponding position such as the portfolio management team. The previously described roles concerning portfolio management can be mapped to the managing layer. Here the daily operational decisions are taken and also monitoring and controlling is located. However, in portfolio management also the steering layer plays an important role as here the guidance and overall responsibilities concerning portfolio governance are located. Having mapped the roles onto the layers subsequently the necessary competencies are mapped onto the roles. In ArchiMate these terms are described as Business role and Capability respectively. This concludes the construction of the reference architecture. An example provided by Cordeiro et. al. [5] makes the theoretical propositions more applicable. In fig. 3.10 the comparative analysis between an example enterprise and the proposed reference architecture is depicted.

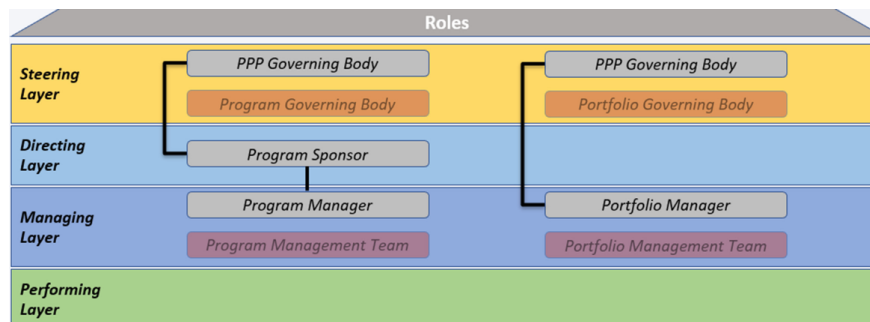


Figure 3.10: Layers Example [5]

Within the portfolio ecosystem one can find that there are overall less roles dedicated to portfolio management. In the steering layer the whole Portfolio Governing Body is missing making the overall PPP governing body responsible for all steering decisions being made in each project, program and portfolio. This in turn leads to the thought that completely informed decisions cannot be taken as the detailed knowledge is most likely not present. Another major difference which can be seen is the missing Portfolio Management Team, in the managing Layer only a Portfolio Manager is present. This means that one portfolio manager must have all competencies needed to fulfill a certain project. In turn, one can conclude that the likelihood of this person being overloaded is significantly elevated comparing the work load to an enterprise where these responsibilities are distributed across a whole team. Although not represented in the example provided, it might also be the case that roles are assigned which are not necessary in an organization leading to an unnecessary overhead. Despite the theoretical framework providing clear

guidelines, one still has to take each organization as a individual one as some especially smaller enterprises might not need some of the proposed roles.

3.3 Integrating Enterprise Architecture and Portfolio Management

So far, the described initiatives to integrate enterprise architecture and portfolio management have been unilateral, while a bilateral integration has not been considered. Such a bilateral approach has only been initiated by two papers found in literature, both of them in the field of IT.

3.3.1 IT Project Alignment

Simon et. al. [41] firstly combined EA and PM to create an integrated perspective of both managing practices. However, the proposed integration cannot be thought of as interlocking gears, but rather a structured and thought through approach to effectively combine enterprise architecture and portfolio management. The integration should maximize their synergies and provide a holistic framework. Enterprise Architecture and their models serve as a basis reasoned by the crucial understanding of dependencies within the portfolio. Through the projects and programs overseen by IT portfolio management, EA offers a foundation for creating the organization's future state. In fact, many components which can be allocated to IT portfolio management like application systems, infrastructure components, IT service and the corresponding projects are already regularly included in EA models.

Figure 3.11 depicts the integration of EA and PM within the main processes applied in enterprises.

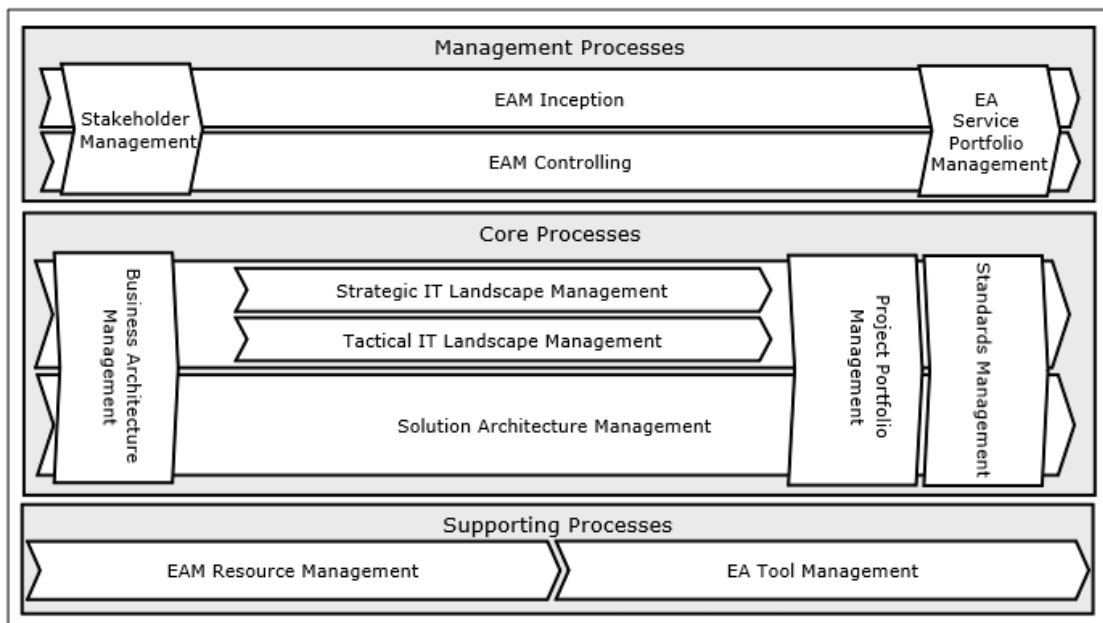


Figure 3.11: EA management and portfolio management integration [41]

Core processes are the primary activities of the organization and, consequently, are a main focus of both, EA and PM. In the context of IT, the corresponding landscape management is an elementary process, as represented as strategic and tactical IT landscape management, in fig. 3.11 within the core process. The former describes the long-term components while the latter target possible future components and technologies which might only be needed for very specific projects. Application Portfolio Management (APM) can be assigned to landscape management as APM focuses on the management of applications needed for the processes within the organization. However, APM has a narrower focus as it addresses the specific use of the applications within the business processes rather than all applications used throughout the entire enterprise. The valuation of the components within the application portfolio can and potentially should be further supported by an analysis regarding potential problems with the functionality and overall health of technology. When assessing the need for new technologies within the context of tactical IT Landscape Management, detailed data by APM needs to be provided for reasonable decision making. Operational activities are encapsulated within Solution Architecture Management, while Business Architecture Management and Standards Management encompass these levels. Simon et. al. [41] incorporated Project Portfolio Management into the core processes, additionally spanning IT landscape management and Solution Architecture Management. Furthermore, the field of process management include the application of several methodologies to detect, evaluate, and enhance an organization's current workflows. Additionally, it includes the management of the services provided. Finally, supporting processes, serve to support the core processes and include activities such as resource management, budgeting and staffing.

If new projects should be added, roadmaps which plan the integration are often used in practice. Over two thirds of the organizations describe the definition of those roadmaps as a task provided by EA. Less than one fifth attribute this responsibility to PM and even fewer regard it as a joint activity. Organizations understanding this process as a joint activity possess the largest IT departments on average. Nevertheless, the process from project approval to its integration should be overlooked by both managing practices. As soon as a project is approved, architectural compliance is evaluated. Consequently, EA management needs to be informed, if the decision for the project has not been done jointly, and the corresponding EA models might to be adapted. If there are issues regarding already started or planned projects, PM must be consolidated to efficiently close possible gaps. Furthermore, project prioritization has to take place. This however, mainly lays in the scope of EA as their models best represent interdependencies, needing even more profound cooperation.

In the study conducted by Simon et. al. [41], the majority of enterprises reported the use of software tool support for either EA, PPM or both. However, two-thirds of the enterprises have not integrated their systems and even less have shared interfaces. Notably, only one enterprise participating in the study uses the same software tool for both, EA and PPM. This lack of integration leads to the conclusion, that the available software tools do not support the need for both managing practices in the needed detail Also, shared

interfaces are rare or do not provide the needed support. However, the study does not evaluate the underlying reasons in detail, instead, it merely suggests a limited integration capacity provided by those tools.

3.3.2 IT Portfolio Management

Although the usefulness to integrate EA and PM has already been recognized in 2013 as described in the previous section, no significant improvements seemed to be made in this area until 2020. This becomes evident by the work of Gellweiler [11] who again thoroughly evaluates the necessity and possible major advantages of integrating the two managing techniques. Gellweiler [11], however, criticises the work done by Simon et. al. [41] because of the definition of the scope of EA and PPM used by the latter as they assigned application systems, infrastructure components and IT services to Portfolio Management. Gellweiler [11], however, recognizes IT being in the application area of EA, while PPM encompasses a broader scope.

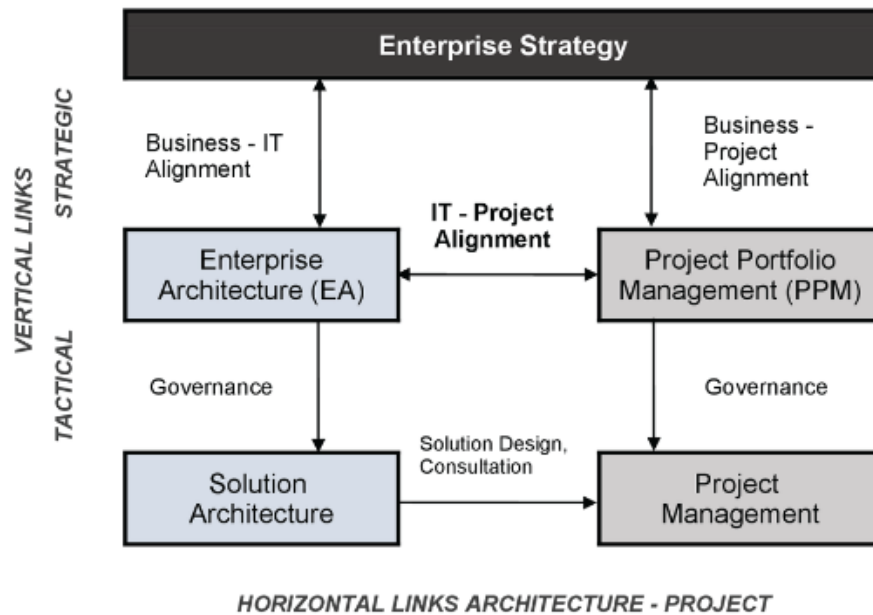


Figure 3.12: IT Portfolio Management [11]

The primary reason for the need to connect the two managing techniques are described as their respective focuses and the connections between them, whereby a change in one of them has to lead to a change in the other one. For example, if a new project is decided for in the PPM scope, naturally, the IT systems will be affected governed by EA. Similarly, the other way around, the introduction of a new technology within the EA domain, will impact at least some components of the project portfolio. Furthermore, managing the requirements an enterprise has from an EA based or PPM based view calls for the integration of the two managing techniques. ArchiMate was proposed as a modeling language for this purpose, as it is the primary modeling language used in

EA, and concepts needed in PM can be mapped to this language as well. This reciprocal relationship emphasizes how managing EA and PPM together is essential to ensuring alignment throughout the enterprise. The relationship proposed is depicted in fig. 3.12. Attention should be paid to the horizontal arrangement of EA and PPM which represents the necessary alignment of the two managing techniques to emphasise the selection and prioritization of the projects that align with the strategic objectives and timelines of the organization.

This objective should be accomplished by utilizing a roadmapping technique which is schematically depicted in fig. 3.13.

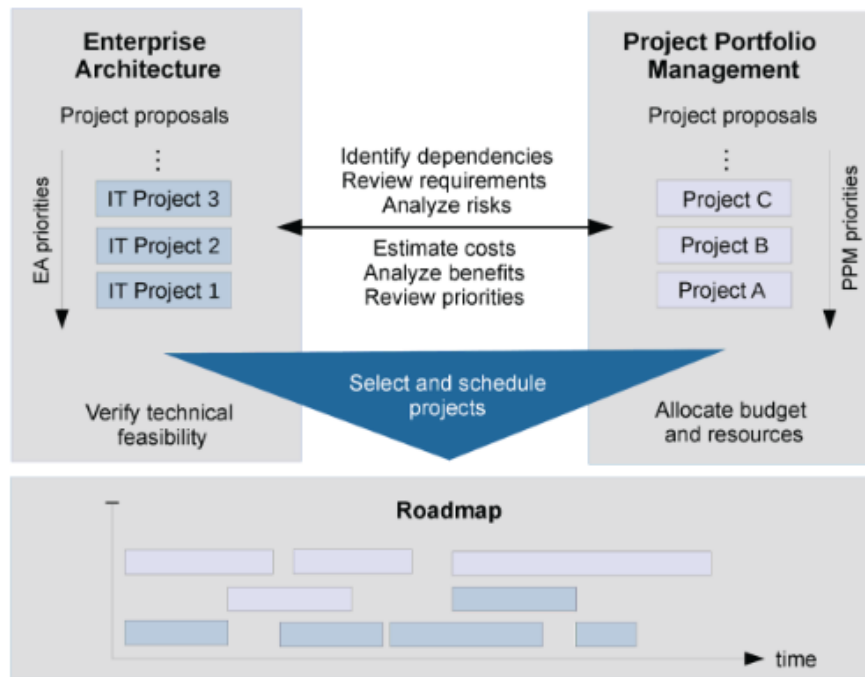


Figure 3.13: IT Portfolio Management Roadmap [11]

This roadmap covers the plans of both EA, and PPM and integrate them into a joint plan. From an EA perspective, roadmaps represent the current state and help to guide towards a future architecture. From the perspective of PPM, the corresponding projects are evaluated regarding their risks and dependencies while a clear association with the overall strategic plan has to be ensured. Both EA and PPM can manage their schedules individually at first and they will consequently be matched. This means, that possible constraints must be discussed, while parts only affecting one part have to be considered in the individual plan, but not brought into the overall schedule discussion. Furthermore, matching the proposed components of each divisions to the overall strategy should not be necessary, as they have to be in line with the enterprises overall strategy. Altogether, the objective of roadmaps is, to ensure a transparent planning process traceable throughout various departments, leading to a cohesive project plan covering all necessary aspects of

both, EA and PPM. The result of this process is a clear plan with a connected objective considering all occurring dependencies with a explicit prioritization and a transparent execution strategy of the projects considering a defined timeline. However, the roadmap is not static. In the event of changes within the enterprise which are common in dynamic environments, the roadmap must be adaptable.

Discussion

Having described the relevant approaches found in literature, the findings should be discussed. By comprehending and linking the approaches, a deeper understanding provided by an overview of the main challenges and how they might be approached should be provided. Furthermore, the practical implications are of interest. Lastly, the limitations encountered during this research are transparently described.

One of the main challenges encountered throughout this thesis is the insufficient, effectively unclear and imprecise definition of the two managing techniques and their corresponding scope of responsibility. This became especially evident in section 3.3, where the only two papers found in literature providing an integrated discussion regarding the linkage between EA and PM, define these concepts completely differently. Having a clear definition would be essential to develop a structured plan to integrate EA and PM, thereby enabling enterprises to optimize the benefits these managing techniques offer.

Although roadmaps have been described, the shared IT infrastructure remains an area requiring further development. Another major challenge is, that approaches are gradually being developed within scientific areas, however, the integration into practical applications in enterprises remains challenging. Even if there is a desire to implement effective communication and cooperation, the lack of practical guidance in literature remains a constraint enterprises have to face. Of course, the described roadmaps can be used as a framework for efficient collaboration. Defining the necessary steps to reach the shared goal of providing a holistic overview can serve as an appropriate starting point for numerous enterprises which have not yet been able to integrate substantial cooperation between the divisions. However, a real connection through unified channels of communication, software tools and also a mutual understanding is still not adequately addressed in literature, and consequently also not applicable in real-world scenarios. This, in fact, can hold decision-makers back from pushing forward with the integration efforts. Nevertheless, some approaches, even though only described from either the perspective of EA or PM, can serve as a beneficial starting point. Especially enterprises that are at the initial

stages of introducing either managing technique, although most likely EA will be newly introduced given the established nature of PM, can generate advantages by taking the influence of the complementary technique on their overall strategy into account.

The limitations of the thesis include the selected languages as well as the choice made in terms of the databases the research is based on. However, as English is the main language used in science, the number of relevant research not taken into consideration is expected to be vanishingly small. Additionally, the specific databases were selected because they cover the most important and recognized research papers, journals and proceedings in the field of both Enterprise Architecture and Portfolio Management. While this might introduce some bias favoring the most widely-cited papers, this approach offers a reliable foundation for this thesis.

Conclusion

In this thesis, a literature study on the definition(s) of Enterprise Architecture and Portfolio Management as well as their interconnectedness reported in literature was conducted. The main objective was to identify the efforts already taken to integrate EA and PM serving to provide a holistic overview of the enterprises as well as to generate a understanding of efficiently working towards the shared goals. This analysis revealed that there has been very limited research carried out taking a comprehensive look into the connections between the two managing techniques. The few papers identified in this literature study mainly focus on unilateral connections which were described in the subdomains of Application Portfolio Management, Technology Portfolio Management, Transformation Portfolio Management, Project Portfolio Management and IT Portfolio Management. Bilateral integration has only been considered in the context of IT, with a focus on Project Portfolio Management, and even in this domain, the research is yet to be mature. Independent of the way of connection, there is a lack of real-world integration of the few described possibilities as well. This shortcoming can be attributed to the absence of comprehensive frameworks and interfaces which can be used by both divisions. The low number of papers in this field put to show the early stage of research within this area.

Future research might focus on looking into further evaluating the given approaches in greater detail, providing a comprehensive and practical approach for implementing these possibilities in practical contexts. Furthermore, as the evaluation process of the components within IT portfolios has already been described from both perspectives in literature, an evident next step would be to evaluate the possibilities of combining the approaches. Providing enterprises with a detailed framework should be a central focus regarding future research, facilitating the application of theoretical and literary knowledge in real-world situations. Moreover, also the integration of EA and PM in other subdomains of PM described warrants further research, as collaborative roadmaps have thus far only been characterized within the context of IT.

5. CONCLUSION

Concluding the thesis, there remains a lot of potential for developing comprehensive frameworks and technologies applicable in real-world scenarios. This can provide major improvements for enterprises across all sectors, effectively leveraging the opportunities EA offers, especially when bilateral recognition integration and of EA and PM is considered.

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Bibliography

- [1] Adina Aldea, Maria-Eugenia Iacob, Maya Daneva, and Latifah Helmy Masyhur. Multi-criteria and model-based analysis for project selection: An integration of capability-based planning, project portfolio management and enterprise architecture. In *2019 IEEE 23rd International Enterprise Distributed Object Computing Workshop (EDOCW)*, pages 128–135. IEEE, 2019.
- [2] Gene M. Amdahl, Gerrit A. Blaauw, and Frederick P. Brooks Jr. Architecture of the IBM system/360. *IBM J. Res. Dev.*, 8(2):87–101, 1964.
- [3] Jay Barney. Firm resources and sustained competitive advantage. *Journal of management*, 17(1):99–120, 1991.
- [4] Robert G Cooper, Scott J Edgett, and Elko J Kleinschmidt. Portfolio management: fundamental to new product success. *The PDMA toolbox for new product development*, 1:331–364, 2002.
- [5] Gonçalo Cordeiro, André Vasconcelos, and Bruno Fragoso. Project, program, portfolio governance model reference architecture in the classic approach to project management. In *International Conference on Enterprise Information Systems*, pages 619–630, 2020.
- [6] Cambridge Dictionary. *technology*. 5 2024.
- [7] DIN. DIN 69909: 1 Multiprojektmanagement–Management von Projektportfolios, Programmen und Projekten–Teil 1: Grundlagen, 2013.
- [8] Mahmoud Ershadi, Marcus Jefferies, Peter Davis, and Mohammad Mojtahedi. Towards successful establishment of a project portfolio management system: business process management approach. *The Journal of Modern Project Management*, 8(1), 2020.
- [9] Matthias Frenzel and Christian Höllwieser. Technologieportfolio-management. *Benutzerzentrierte Unternehmensarchitekturen: Ein portfolio-orientierter Ansatz zur Geschäftstransformation mit ArchiMate®*, pages 179–206, 2020.
- [10] Gartner Group. Best Enterprise Architecture Tools Reviews 2024 | Gartner Peer Insights.

- [11] Christof Gellweiler. Connecting enterprise architecture and project portfolio management: A review and a model for it project alignment. *International Journal of Information Technology Project Management (IJITPM)*, 11(1):99–114, 2020.
- [12] Sebastian Godenhjelm, Rolf A Lundin, and Stefan Sjöblom. Projectification in the public sector—the case of the european union. *International Journal of Managing Projects in Business*, 8(2):324–348, 2015.
- [13] Danny Greefhorst and Erik Proper. *Architecture Principles - The Cornerstones of Enterprise Architecture*, volume 4 of *The Enterprise Engineering Series*. Springer, 2011.
- [14] Gartner Group. Enterprise Architecture (EA), 2024.
- [15] Hong Guo, Jingyue Li, Shang Gao, and Darja Smite. Agile enterprise architecture by leveraging use cases. In *Proceedings of the 16th International Conference on Evaluation of Novel Approaches to Software Engineering (ENASE 2021)*. SciTePress, 2021.
- [16] Kazem Haki and Christine Legner. The mechanics of enterprise architecture principles. *Journal of the Association for Information Systems*, 22(5):1334–1375, 2021.
- [17] Lars Kristian Hansen and Per Svejvig. Seven decades of project portfolio management research (1950–2019) and perspectives for the future. *Project Management Journal*, 53(3):277–294, 2022.
- [18] Infrastructure & Projects Authority. Assurance Workbook. Technical report, 7 2021.
- [19] Project Management Institute. *The Standard for Portfolio Management*. Project Management Inst, 1 2017.
- [20] Michael C. Jensen. Value maximization, stakeholder theory, and the corporate objective function. *Business ethics quarterly*, pages 235–256, 2002.
- [21] Henk Jonkers, Marc M. Lankhorst, René van Buuren, Stijn Hoppenbrouwers, Marcello M. Bonsangue, and Leendert W. N. van der Torre. Concepts for modeling enterprise architectures. *Int. J. Cooperative Inf. Syst.*, 13(3):257–287, 2004.
- [22] Dimitris Karagiannis, Christoph Moser, and Anke Helmes. Einleitung. *Benutzerzentrierte Unternehmensarchitekturen: Ein portfolio-orientierter Ansatz zur Geschäfts-transformation mit ArchiMate®*, pages 1–12, 2020.
- [23] Vilas Janardan Kharat and Koteswara Rao Naik Bhukya. Best practices in project portfolio management for dynamic decision making. *The Journal of Modern Project Management*, 6(1), 2018.
- [24] Catherine P Killen. Organizational agility through project portfolio management. *Portfolio management: a strategic approach*, 17(1), 2014.

- [25] Lutz Kirchner, Matthias Frenzel, and Christoph Moser. Applikationsportfolio-management. *Benutzerzentrierte Unternehmensarchitekturen: Ein portfolio-orientierter Ansatz zur Geschäftstransformation mit ArchiMate®*, pages 119–148, 2020.
- [26] Radoslaw Klimek and Piotr Szwed. Verification of archimate process specifications based on deductive temporal reasoning. In Maria Ganzha, Leszek A. Maciaszek, and Marcin Paprzycki, editors, *Proceedings of the 2013 Federated Conference on Computer Science and Information Systems, Kraków, Poland, September 8-11, 2013*, pages 1103–1110, 2013.
- [27] Janne J Korhonen and Marco Halén. Enterprise architecture for digital transformation. In *2017 IEEE 19th Conference on Business Informatics (CBI)*, volume 1, pages 349–358. IEEE, 2017.
- [28] Martin Op’t Land and Erik Proper. Impact of principles on enterprise engineering. In Hubert Österle, Joachim Schelp, and Robert Winter, editors, *Proceedings of the Fifteenth European Conference on Information Systems, ECIS 2007, St. Gallen, Switzerland, 2007*, pages 1965–1976. University of St. Gallen, 2007.
- [29] Martin Op’t Land, Erik Proper, Maarten Waage, Jeroen Cloo, and Claudia Steghuis. *Enterprise Architecture - Creating Value by Informed Governance*. The Enterprise Engineering Series. Springer, 2009.
- [30] Marc M. Lankhorst, editor. *Enterprise Architecture at Work - Modelling, Communication and Analysis, Fourth Edition*. Springer, 2017.
- [31] Dennis Lock and Reinhard Wagner. *The handbook of project portfolio management*. Routledge, 2018.
- [32] Miia Martinsuo. Project portfolio management in practice and in context. *International journal of project management*, 31(6):794–803, 2013.
- [33] Sandra Matos and Eurico Lopes. Prince2 or PMBOK—a question of choice. *Procedia Technology*, 9:787–794, 2013.
- [34] Zhang Mengmeng, CHEN Honghui, and Liu Junxian. Resource allocation approach to associate business-it alignment to enterprise architecture design. *Journal of Systems Engineering and Electronics*, 30(2):343–351, 2019.
- [35] Michael Möhring, Barbara Keller, Rainer Schmidt, Kurt Sandkuhl, and Alfred Zimmermann. Digitalization and enterprise architecture management: a perspective on benefits and challenges. *SN business & economics*, 3(2):46, 2023.
- [36] Christoph Moser and Felix Brandmayr. Transformationsportfolio-management. *Benutzerzentrierte Unternehmensarchitekturen: Ein portfolio-orientierter Ansatz zur Geschäftstransformation mit ArchiMate®*, pages 59–89, 2020.

- [37] Dick Quartel, Maarten WA Steen, and Marc M Lankhorst. Application and project portfolio valuation using enterprise architecture and business requirements modelling. *Enterprise Information Systems*, 6(2):189–213, 2012.
- [38] Patrick Saint-Louis, Marcklyvens C Morency, and James Lapalme. Examination of explicit definitions of enterprise architecture. *International Journal of Engineering Business Management*, 11:1847979019866337, 2019.
- [39] Farrukh Saleem and Bahjat Fakieh. Enterprise architecture and organizational benefits: A case study. *Sustainability*, 12(19):8237, 2020.
- [40] Mary Shaw and David Garlan. *Software architecture - perspectives on an emerging discipline*. Prentice Hall, 1996.
- [41] Daniel Simon, Kai Fischbach, and Detlef Schoder. Integrating it portfolio management with enterprise architecture management. *Enterprise Modelling and Information Systems Architectures (EMISAJ)*, 8(2):79–104, 2013.
- [42] IEEE SMC. Enterprise Architecture and Engineering - IEEE SMC, 5 2022.
- [43] Orbus Software. Research detail, 2024.
- [44] Toomas Tamm, Peter B. Seddon, Graeme G. Shanks, and Peter Reynolds. How does enterprise architecture add value to organisations? *Commun. Assoc. Inf. Syst.*, 28:10, 2011.
- [45] The Open Group. TOGAF 9 - The Open Group Architecture Framework Version 9, 2009.
- [46] The Open Group. 9. Application Layer : ArchiMate® 3.1 specification, 2019.
- [47] Reinhard Wagner. *Erfolgreiches Projektportfoliomanagement*. Symposion Publishing GmbH, 2016.
- [48] John A Zachman. The zachman framework for enterprise architecture. *Primer for Enterprise Engineering and Manufacturing.[si]: Zachman International*, 2003.